1. Clinical Question & Core Hypothesis

Clinical Question

The Ontologist highlights that imaging-inferred immune context (in-situ infiltrates/TILs), PD-L1 status, and prostaglandin signaling (PTGER4) co-vary with outcome after therapy — raising the question: why do some TIL-rich, docetaxel-treated high-risk tumors still fail to show improved survival? 2) Tier 1 paths used: I built on Core Path 1 (status of in-situ immune cell infiltrate informs outcome and is captured by imaging/pathomics) and Core Path 3 (PTGER4–PD-L1–T-cell exhaustion axis). These connect a measurable imaging feature (TIL density) to a plausible immune-suppressive mechanism (PTGER4-driven PD-L1/T-cell dysfunction). 3) Tier 2 support: radiomics/imaging biomarkers and PSA nadir provide noninvasive/temporal readouts that can validate the TIL–outcome link; CENPA and SOX2 suggest tumor-intrinsic aggressiveness that could modulate immune effectiveness; CSF1 indicates an inflammatory/cytokine milieu that may reinforce immune suppression. 4) Endpoint choice: Overall survival (OS) is the most robust, clinically definitive endpoint available in the dataset and is the focus of the core paths linking therapy, immune state, and outcome. 5) Deliberate exclusions: I excluded exploratory/neuro/cross-species nodes (SPC, cognitive uptake) and neuroendocrine focus (SOX2 placed only as secondary) to keep the hypothesis mechanistically tight and testable with one imaging readout and two molecular anchors?

Working Hypothesis

High tumor-infiltrating lymphocyte (TIL) density does not predict improved overall survival in docetaxel-treated, high-risk localized prostate cancers when tumor-intrinsic prostaglandin E2 signaling via PTGER4 drives PD-L1–mediated T-cell exhaustion. Consequently, patients whose tumors express high PTGER4 and CD274 within a TIL-rich microenvironment will have worse overall survival and may benefit from combination PTGER4 pathway inhibition and PD-1/PD-L1 blockade to reinvigorate antitumor immunity.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	CD274, PTGER4
Primary endpoint	OS

2. Why This Gene — Not Others

Primary hypothesis gene(s)

CD274: Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

PTGER4: PTGER4) co-vary with outcome after therapy — raising the question: why do some TIL-rich, docetaxel-treated high-risk tumors still fail to show improved survival? 2) Tier 1 paths used: I built on Core Path 1 (status of in-situ immune cell infiltrate informs outcome and is captured by imaging/pathomics) and Core Path 3 (PTGER4–PD-L1–T-cell exhaustion axis).

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	CENPA	Supporting modifier from knowledge graph
Secondary	SOX2	Supporting modifier from knowledge graph
Secondary	CSF1	Supporting modifier from knowledge graph
Secondary	PSA	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: retrospective cohort analyses correlating TIL density (imaging/pathomics) with OS, stratified by PTGER4 and CD274 expression (IHC/RNA-seq) and PSA nadir as a temporal readout.

Prediction 2: In vitro/ex vivo assays using prostate cancer cell lines or organoids with PTGER4 modulation to assess PD-L1 induction and T-cell activity.

Prediction 3: In vivo models to evaluate whether PTGER4 antagonism plus PD-1/PD-L1 blockade improves antitumor immunity in TIL-rich tumors. Expected outcome: a clear subset of patients with TIL-high, PTGER4/CD274-expressing tumors will have worse OS, identifying candidates for combined PTGER4–PD-1/PD-L1 therapies.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Gene expression does not equal protein activity; phospho-protein or functional assays are needed for confirmation.

Alternative Directions

- CENPA: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- SOX2: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.